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A Quick Introduction to Git & GitHub (April 2026)

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GitHub

- **GitHub is a website and cloud-based service that helps developers store and manage their code, as well as track and control changes to their code, among other features.**
- **GitHub started as a control version system but …**

About GitHub

GitHub is a cloud-based platform where you can store, share, and work together with others to write code.

Storing your code in a "repository" on GitHub allows you to:

- **Showcase or share** your work.
- **Track and manage** changes to your code over time.
- Let others **review** your code, and make suggestions to improve it.
- **Collaborate** on a shared project, without worrying that your changes will impact the work of your collaborators before you're ready to integrate them.

Collaborative working, one of GitHub's fundamental features, is made possible by the open-source software, Git, upon which GitHub is built.

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What is version control?

- Version control helps developers track and manage changes to a software project's code. As a software project grows, version control becomes essential.
- Version control lets developers safely work collaboratively either simultaneously or asynchronously.
- All of these changes are then tracked and can be reverted if need be.

What is Git?

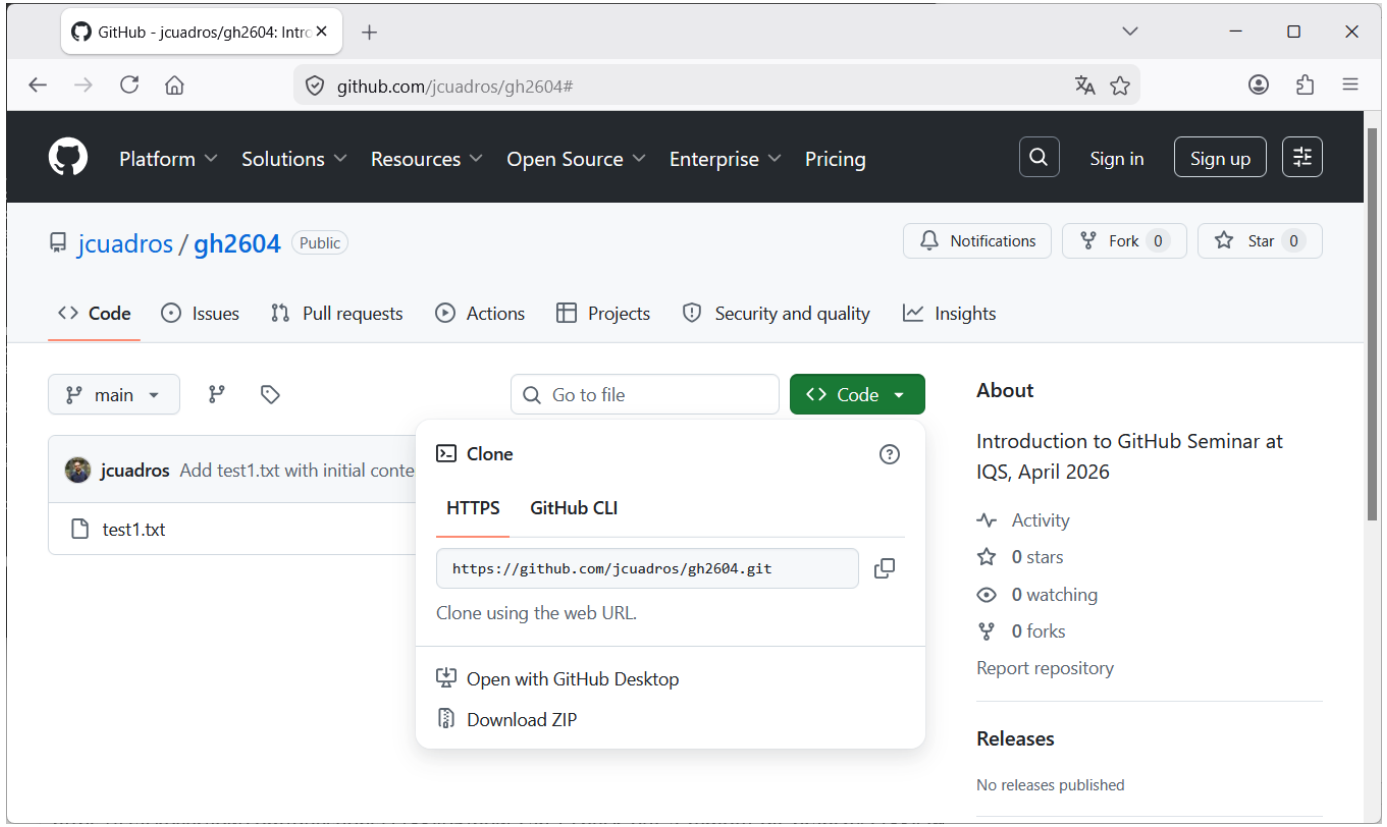
- Git is a specific open-source version control system created by Linus Torvalds in 2005.
- Specifically, Git is a distributed version control system, which means that the entire codebase and history is available on every developer's computer, which allows for easy branching and merging.

Before starting

- **Create an account on GitHub**
  - https://github.com/
- **Install Git**
  - https://git-scm.com/

Let’s start

- **Let’s go to my GitHub account**
  - https://github.com/jcuadros
  - https://github.com/jcuadros/gh2604



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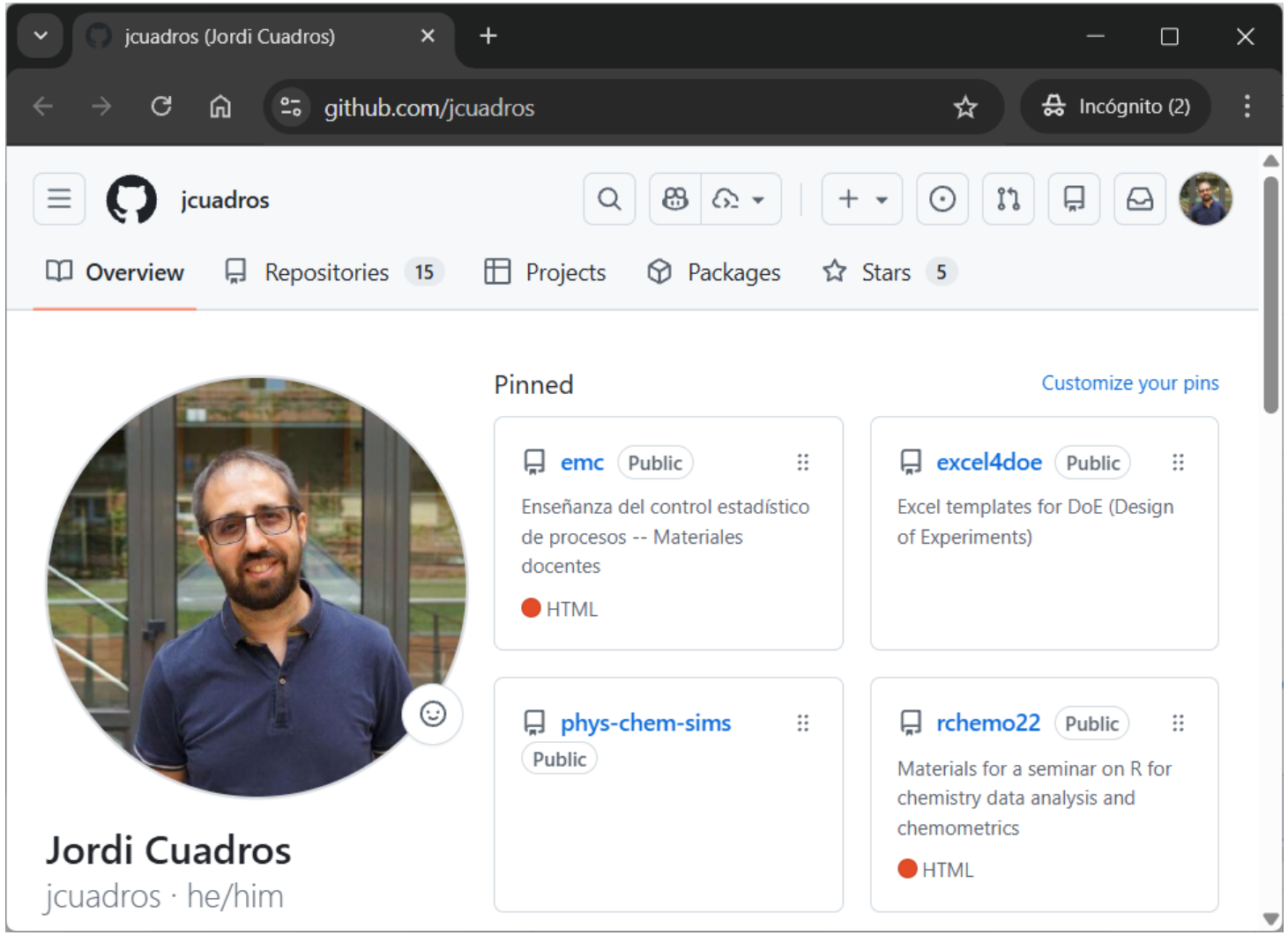
GitHub/Git interfaces

- **GitHub web interface**
- **Git CLI**
- **GitHub desktop**
  - https://github.com/apps/desktop
- **TortoiseGit**
  - https://tortoisegit.org/
- **Any programming editor (VSCode, RStudio…)**

GitHub web interface

- Create a repo
- Add a file
- Commit
- Check history
- Add contributors



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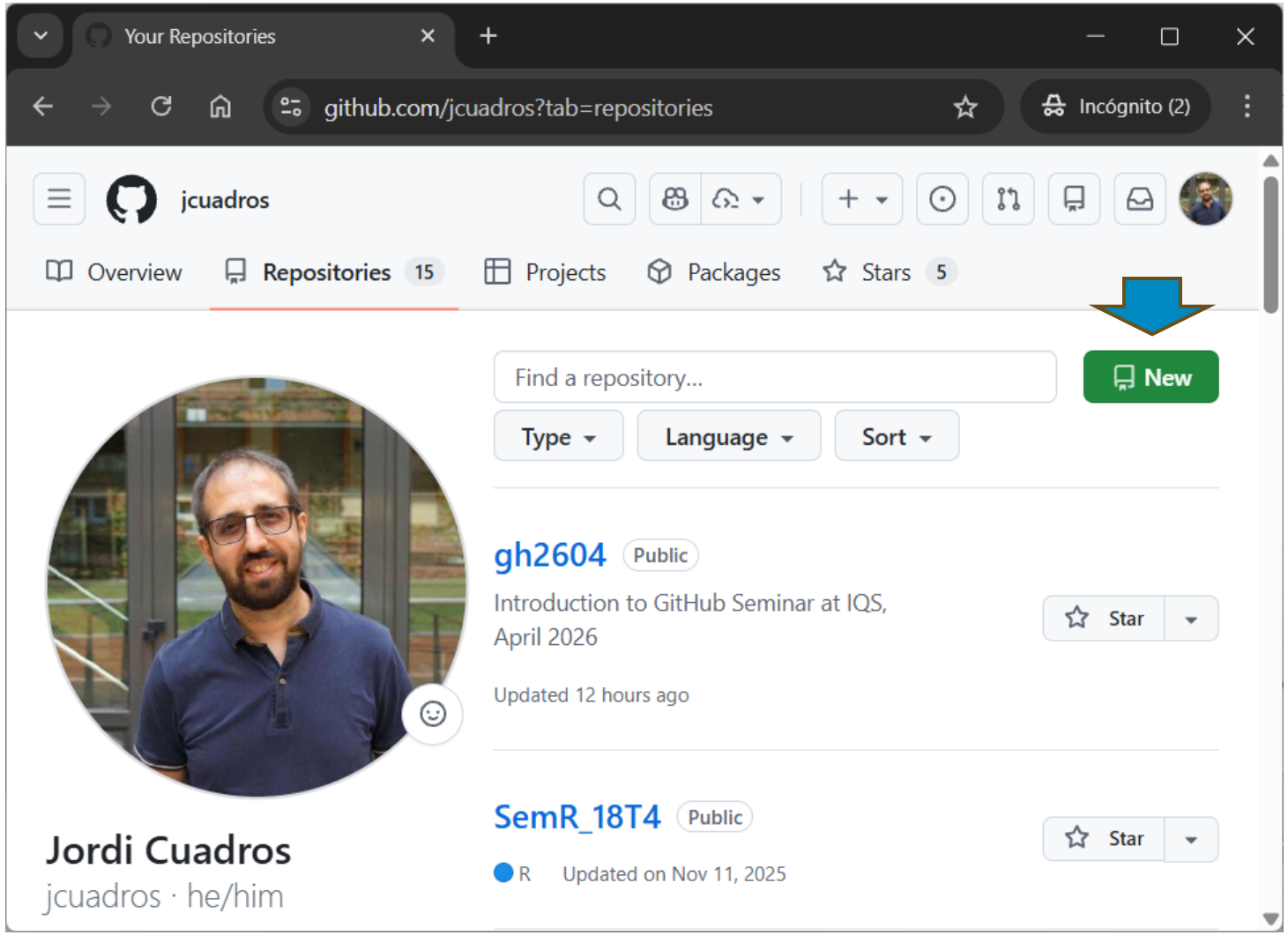

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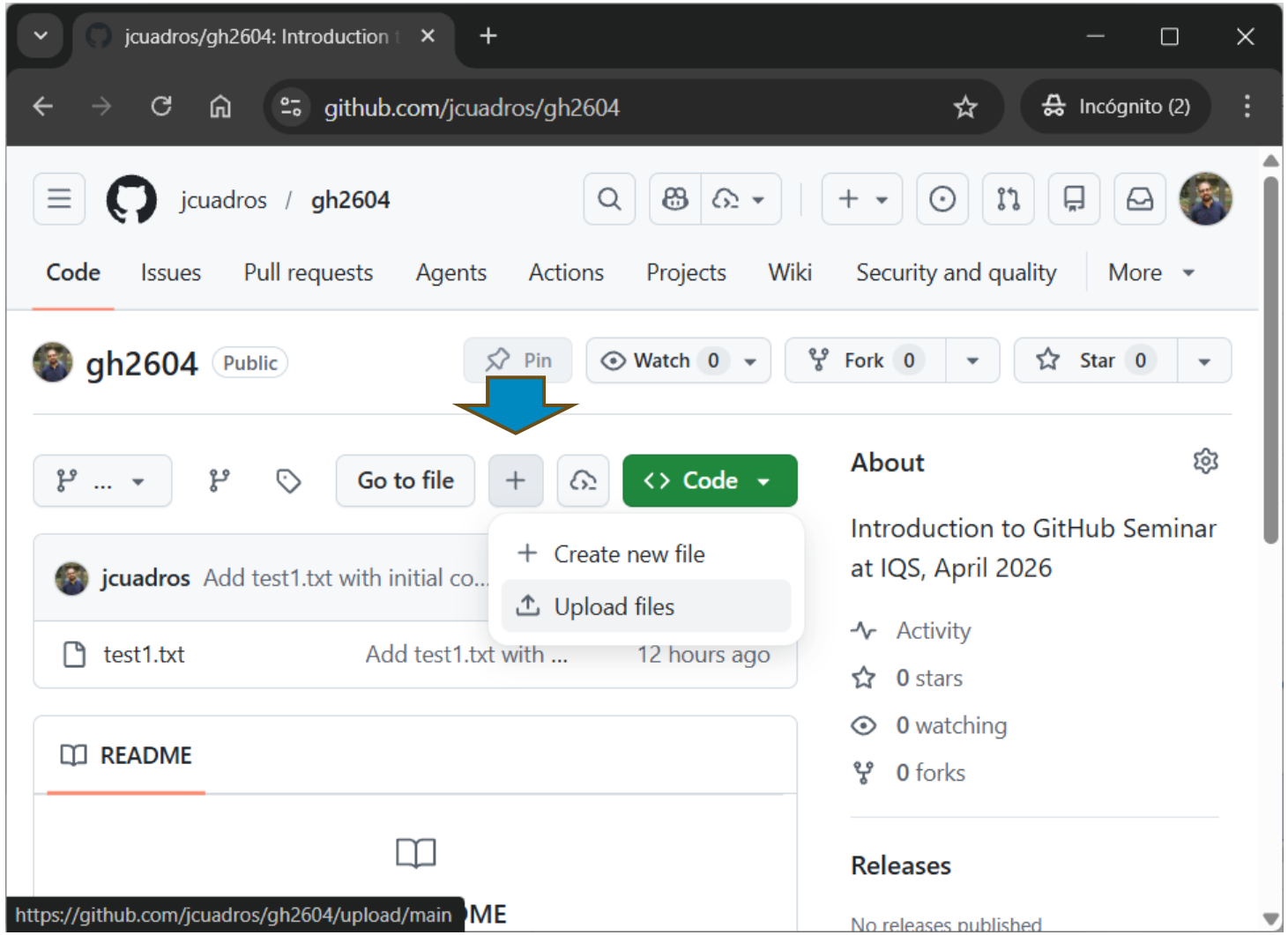

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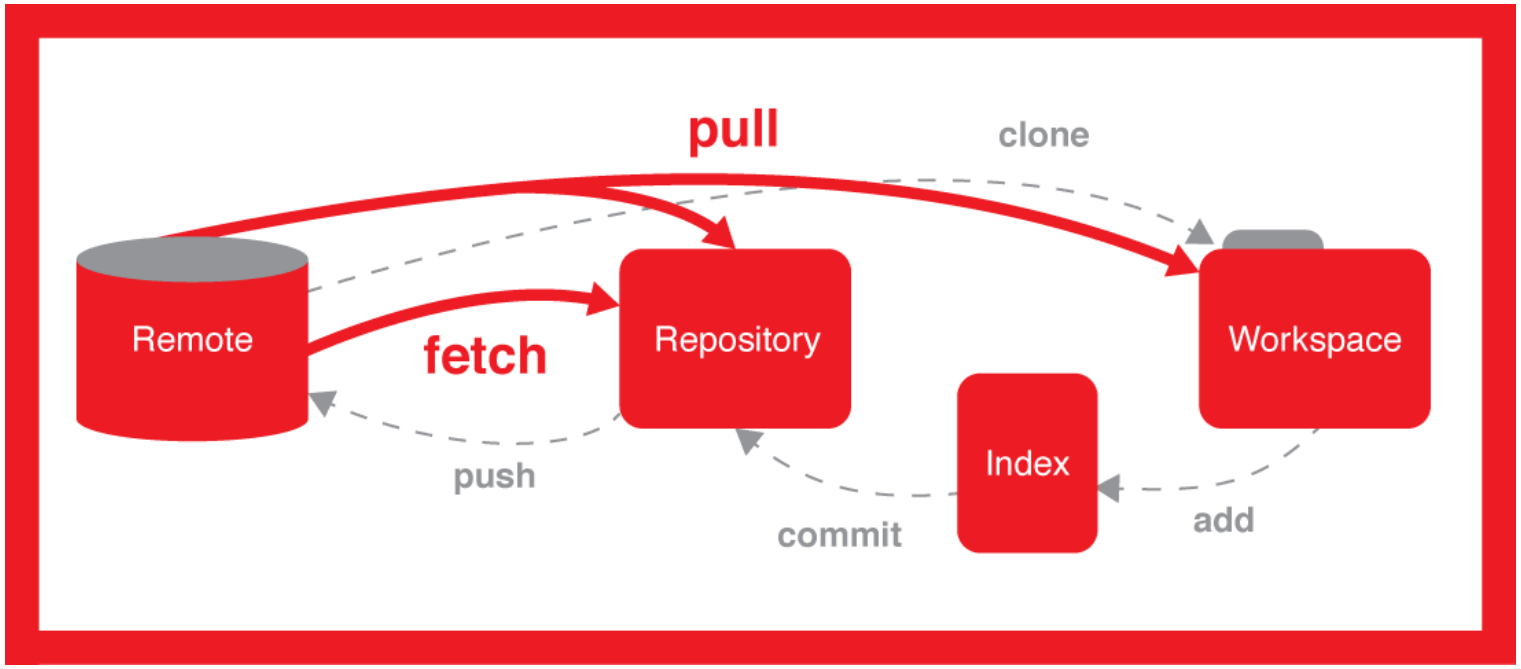
Git CLI

- **Git Command Line Interface**
  - https://git-scm.com/cheat-sheet

Git simplest workflow

- Create your repository
- Clone
- Edit
- Commit
- Pull
- Push (requires editing rights)



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Conflicts resolution

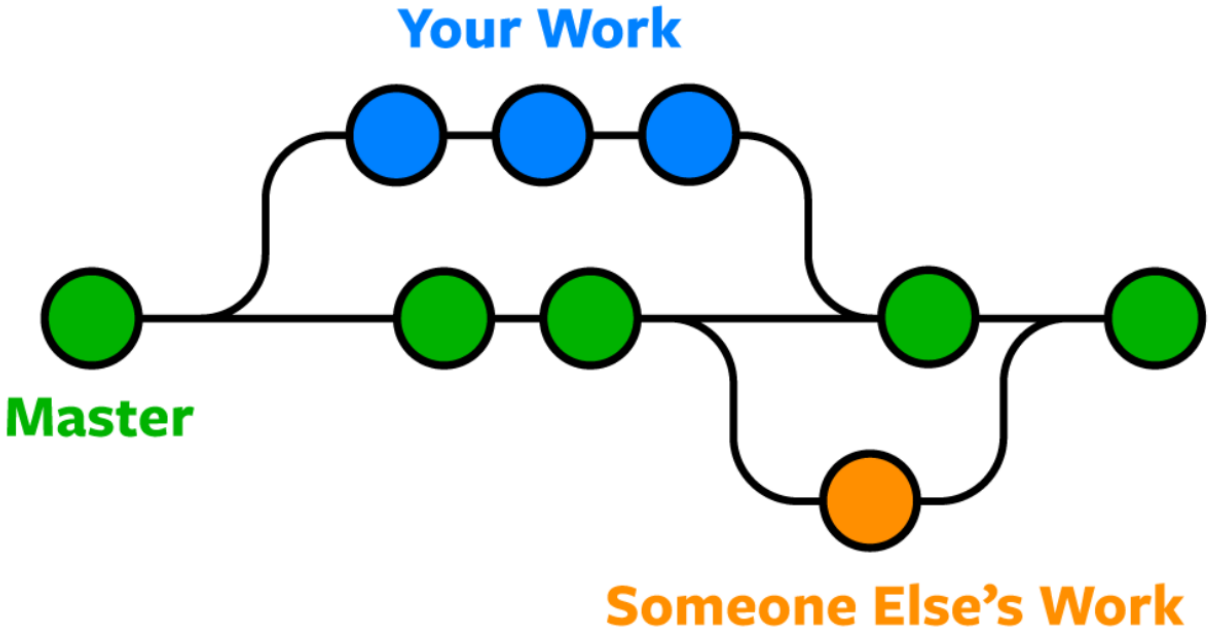
- **If the changes in the repository and your changes are not compatible, there is a CONFLICT**
- **In this case**
  - Commit
  - Pull (as standard)
  - Solve the conflict
  - Commit
  - Pull (again)
  - Push

Working with branches and forks

- **Download**
- **Clone**
- **Fork**
  - Closes with a pull request

Branching and merging

- **Branch**
- **Closing**
  - With merge
  - Through a pull request



GitHub campus

Other features

- GitHub Codespaces
- GitHub Projects
- GitHub Teams
- GitHub Copilot
- GitHub Docs
- GitHub Issues
- GitHub Discussions

Using .gitignore

- The .gitignore files allows specifying which will not be committed and pushed to the repository

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